W UNIVERSITY of WASHINGTON

**UW Math AI Lab**
Spring 2026 · Provable Computation

VERIFIED EXECUTABLE LINEAR ALGEBRA

# DenseMatrix makes Lean matrix algorithms run like data.

Row-major data for RREF, LU, and determinant kernels, bridged back to proof-friendly Matrix theorems.


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SCAN FOR ARTIFACT

## Why DenseMatrix

Row-major arrays let kernels work over contiguous data instead of re-evaluating pointwise Matrix functions.

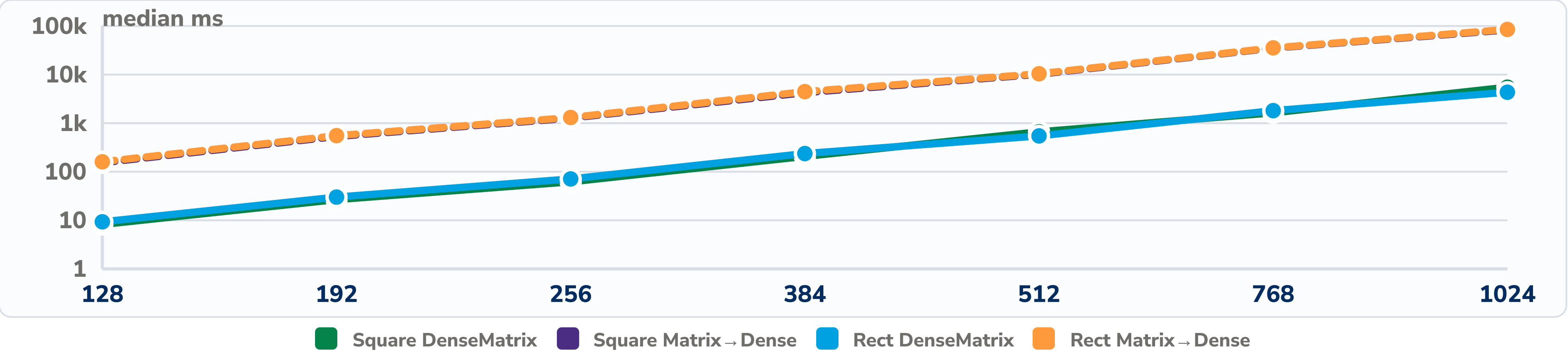
Matrix API → DenseMatrix.ofMatrix → Row-major kernels

Materialized executable workloads only; Matrix remains the proof API.

## Multiplication Scaling

**15.1-21.5x faster**

DenseMatrix vs Matrix→Dense, median ms on a log scale.




## Benchmark Scope

**2247** RECORDS

**328** MATERIALIZED CASES

**2048** MAX ROWS

| | |
|---|---|
| **Protocol** | 30 repeats · 5 warmups |
| **Sizes** | 16, 32, 64, 96, 128, 192, 256 |
| **Stress** | 384, 512, 768, 1024 |
| **Machine** | pinned core · environment archived |

## Proof Connection

- `DenseMatrix.toMatrix_add`
- `DenseMatrix.toMatrix_mul`
- `rowEchelonForm_toMatrix`
- `luDet_eq_det`
- `gaussDet_eq_det`

Theorems connect executable kernels back to the Matrix API.

## Representation Tradeoff

Median ms for pointwise primitives at n=1024.

| OPERATION | DENSE | MATRIX | MATRIX→DENSE | FASTEST |
|---|---|---|---|---|
| **Add** | 54.0 | 45.1 | 47.5 | Matrix |
| **Scalar** | 53.2 | 30.3 | 35.3 | Matrix |
| **Transpose** | 55.3 | 43.5 | 46.8 | Matrix |

## Algorithm Kernels

Representative Rat kernels at n=128.

| | |
|---|---|
| **REF** | 16292 ms |
| **RREF** | 32394 ms |
| **LU factor** | 17090 ms |
| **Gauss det** | 16234 ms |
| **LU det** | 16933 ms |